

9.6.4

A camera shop stocks eight different types of batteries, one of which is type A76. Assume there are at least 30 batteries of each type.

- a. How many ways can a total inventory of 30 batteries be distributed among the eight different types?
- b. How many ways can a total inventory of 30 batteries be distributed among the eight different types if the inventory must include at least four A76 batteries?
- c. How many ways can a total inventory of 30 batteries be distributed among the eight different types if the inventory includes at most three A76 batteries.

- a. How many ways can a total inventory of 30 batteries be distributed among the eight different types?

There're eight different types of batteries, and 30 batteries in total. Since the order does not matter, and repetition is allowed, the number of ways the total inventory of 30 batteries can be distributed among the eight different types is:

$$\binom{n+r-1}{r}$$

Let n be 8, and let r be 30. Then:

$$\binom{8+30-1}{30} = \binom{37}{30} = \frac{37!}{30!(37-30)!} = \frac{37!}{30!7!} = \boxed{10,295,472}$$

Solution: a. There're 10,295,472 number of ways the total inventory of 30 batteries can be distributed among the eight different types.

- b. How many ways can a total inventory of 30 batteries be distributed among the eight different types if the inventory must include at least four A76 batteries?

There're eight different types of batteries, and 30 batteries in total. Since we now must include at least four A76 batteries, we can reserve those slots, and then freely distribute the rest. Hence, the number of ways the batteries can be distributed among the eight different types is:

$$\binom{8+26-1}{26} = \binom{33}{26} = \frac{33!}{26!(33-26)!} = \frac{33!}{26!7!} = \boxed{4,272,048}$$

Solution: b. There're 4,272,048 number of ways the total inventory of 30 batteries can be distributed among the eight different types, with at least four A76 batteries.

c. How many ways can a total inventory of 30 batteries be distributed among the eight different types if the inventory includes at most three A76 batteries.

Let S be the total number of ways a total inventory of 30 batteries can be distributed among eight different types.

$$N(S) = \frac{37!}{30!7!} = 10,295,472$$

And let A be the total number of ways a total inventory of 30 batteries can be distributed among eight different types, with at least four A76 batteries.

$$N(A) = \frac{33!}{26!7!} = 4,272,048$$

Then the number of ways a total inventory of 30 batteries can be distributed among the eight different types, if the inventory includes at most three A76 batteries is:

$$N(S) - N(A) = \boxed{6,023,424}$$

Solution: c. There're 6,023,424 number of ways the total inventory of 30 batteries can be distributed among the eight different types, if the inventory includes at most three A76 batteries.